

Life Music: The Sonification of Proteins

John Dunn
and Mary Anne Clark

BIOLOGIST MARY ANNE CLARK: MUSIC AND PROTEINS

I love to walk into the music building at Texas Wesleyan University, which is next door to the science building. Through the doors of the practice rooms, I can hear fragments of 1,000 years of written music, played or sung by the current generation of music students—some with finesse, some with hesitation, some with wild improvisation. I think that if somehow I could walk into a living cell, I would hear something similar—the ribosomes ticking away at the synthesis of proteins, playing out their amino acid sequences, note by note, according to a genetic score that is reproduced sometimes with utter fidelity, sometimes with a few unscheduled substitutions and sometimes with stunningly inventive flourishes. Every generation of cells in every living organism plays the genetic score of its species. However, while the history of music as we know it goes back some 1,000 years, the history of genetic music has been at least 3.8 billion years in the making.

Over a decade ago, I went to a faculty seminar to hear a colleague talk about musical composition. As he discussed how he went about selecting, modifying and organizing musical themes, I was struck by the parallels between musical structure and the structure of proteins and the genes that encode them. Proteins also seem to be composed of phrases organized into themes. For years, I was haunted by this idea and tried occasionally to interest musicians in making the transformation for me—converting a protein sequence into a musical sequence.

I was convinced that this would be worth doing—that amino acid sequences would have the right balance of complexity and patterning to generate musical combinations that would be both aesthetically interesting and biologically informative. There are 20 amino acids in proteins (Table 1), enough to cover a span of about three octaves of a diatonic scale. The amino acids are not arranged at random, just as notes are not arranged at random in a piece of music. Both proteins and music are meaningful. The meaning of a protein is its function in the organism.

For example, the protein hemoglobin serves the function of oxygen binding. Figure 1 represents the sequence of beta globin, which forms half of hemoglobin. Some features of the hemoglobin sequence, or "tune," can be seen by comparing the proteins of different species, which play this tune as variations on a theme. For example, the tuatara, an exotic three-eyed lizard, would seem to have little in common with humans, but the similarities between the human and tuatara beta globin sequences indicate that both proteins are variations on a theme that was in existence before the divergence

of the mammalian and reptilian lineages 200 million years ago. Other variations of beta globin can be found in vertebrate species from all over the world, e.g. Australian ghost bats, Brazilian tapirs, Kenyan clawed frogs, Antarctic dragon fish, and Emperor penguins. Although the beta globin sequences are not identical in these species, they are similar enough that, if converted to music, they would be recognizable as variations on a common theme.

While it seemed obvious to me that proteins had an inherently musical structure, I did not hear a musical translation of a protein until 1996. In the process of preparing to teach a course on structural similarities between proteins and music, I conducted a search via the Internet for others who might also be interested in these parallels (see the Appendix for a current resource list). On John Dunn's algorithmic music site, I found both music based on DNA and protein sequences and the software that would make the musical translation. I purchased one of the software programs to use in the class and discovered that proteins were even more musical than I had anticipated.

ARTIST JOHN DUNN: NATURE AS A TEMPLATE FOR ART MUSIC

An artist working in the medium of sound is liberated from some of the cultural imperatives imposed by traditional music, but at a high cost. Music of all cultures is rich in tradition and convention. Not only do listeners expect to hear musical references they have become familiar with, but cultural and musical tradition also gives music its deep structure. This deep structure is not heard at the conscious level by most listeners, but it is an essential component of any musical work: it is the component that keeps our interest fresh on repeated hearings. Popular music depends on extra-musical cultural associations to accomplish this to a large degree, and so in a rapidly evolving culture popular music must be remade con-

ABSTRACT

An artist and a biologist have collaborated on the sonification of protein data to produce the audio compact disc *Life Music*. Here they describe the process by which this collaboration merges scientific knowledge and artistic expression to produce soundscapes from the basic building blocks of life. The soundscapes may be encountered as aesthetic experiences, as scientific inquiries or as both. The authors describe the rationale both for the artistic use of science and for the scientific use of art from the separate viewpoints of artist and scientist.

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Table 1. Twenty amino acids, their single-letter data-base codes (SLC) and their corresponding DNA codons [6,7]. In this table, the 20 amino acids found in proteins are listed, along with the single-letter codes used to represent these amino acids in protein databases. The DNA codons representing each amino acid are also listed. All 64 possible three-letter combinations of the DNA coding units T, C, A and G are used either to encode one of these amino acids or as one of the three stop codons that signals the end of a sequence. While DNA can be decoded unambiguously, it is not possible to predict a DNA sequence from its protein sequence. Because most amino acids have multiple codons, a number of possible DNA sequences might represent the same protein sequence.

Amino acid	SLC	DNA codons
Isoleucine	I	ATT, ATC, ATA
Leucine	L	CTT, CTC, CTA, CTG, TTA, TTG
Valine	V	GTT, GTC, GTA, GTG
Phenylalanine	F	TTT, TTC
Methionine	M	ATG
Cysteine	C	TGT, TGC
Alanine	A	GCT, GCC, GCA, GCG
Glycine	G	GGT, GGC, GGA, GGG
Proline	P	CCT, CCC, CCA, CCG
Threonine	T	ACT, ACC, ACA, ACG
Serine	S	TCT, TCC, TCA, TCG, AGT, AGC
Tyrosine	T	TAT, TAC
Tryptophan	W	TGG
Glutamine	Q	CAA, CAG
Asparagine	N	AAT, AAC
Histidine	H	CAT, CAC
Glutamic acid	E	GAA, GAG
Aspartic acid	D	GAT, GAC
Lysine	K	AAA, AAG
Arginine	R	CGT, CGC, CGA, CGG, AGA, AGG
Stop codons	Stop	TAA, TAG, TGA

stantly. Classical concert and liturgical music depend far more on multiple layers of abstraction within the music itself, with cultural traditions of harmony and melody that evolve slowly. There are extra-musical associations in classical music, to be sure, but the primary deep structure lies within the music itself; thus, classical music stays fresh in our ears even over centuries.

Midway into the twentieth century, when electronics in general and the tape recorder in particular opened vast

landscapes of tonal colors and compositional layering to musical explorers, it quickly became apparent that no one outside of the electronic music community was listening. Most people considered “electronic music” to be an oxymoron. The problem was not that the electronically generated tones were uninteresting. The problem was that there existed no deep structure in the music, either internally or culturally.

As an early experimenter with electronic music, starting in the 1960s with

multiple tape decks and a razor blade—musique concrète, it was called then—I vividly remember my first hearing of Carlos’s *Switched on Bach* [1], the first electronic music recording to receive popular acclaim. I was driving at the time and nearly ran my car off the road. This was astounding: pure synthesized music that made no attempt whatsoever to mimic conventional instrumentation, that stood on its own as music. Until then, electronic music, even my own—especially my own—had been of interest only because it was electronic and experimental. It had little to do with music as an aesthetic experience and it rarely got a second hearing.

The fly in the ointment, of course, was that the structure that gave *Switched on Bach* its meaning was borrowed from Bach and from the vast tradition of Western harmony. In the end this was imitative music after all, barely hinting at the new musical landscapes that had opened up to electronic composers. Morton Subotnick, arguably the best of the early composers of abstract synthesized electronic music, with several landmark albums to his credit, remarked when asked what kind of music he listened to that he preferred Mozart, Bach and other traditional Western composers. He pointed out that electronic music had no history, no tradition and thus, for the present, little to hold a listener’s interest [2].

Early on, I had determined that my path to composing electronic music would eschew traditional composition and that I would treat this new medium as a separate art form: sound as an artist’s medium, rather than music as a traditionally trained musician would approach it. The reason for this, to me, was obvious. The great investment of tra-

Human:	VHLTP	EEKSA	VTALW	GKVVN	DEVGG	EALGR	LLVVY	PWTQR	FFESF	GDLST
Tuatara:	VHWT	EEKQL	VTSLW	TKVVN	DECGG	EALGR	LLIVY	PWTQR	FFSSF	GNLSS
	PDAVM	GNPKV	KAHGK	KVLGA	FSDGL	AHLDN	LKGTf	ATLSE	LHCDK	LHVDP
	STAIC	GNPRV	KAHGK	KVFTS	FGEAV	KNLDN	IKATY	AKLSE	LHCEK	LHVDP
	ENFRL	LGNVL	VCVLA	HHFGK	EFTPP	VQAAY	QKVVA	GVANA	LAHKY	H
	QNFNL	LGDIF	IIVLA	AHFGK	DFTPA	CQAAW	QKLVR	VVAHA	LAYHY	H

Fig. 1. Beta globins: Comparison of human and tuatara sequences [8]. These two sequences are the single-letter database codes for the amino-acid sequence of the protein beta globin in two species: human and tuatara (a primitive, lizard-like reptile). In each double row of letters, the human sequence is printed above the dividing line, and the tuatara sequence below the line. The letters of the two sequences have been separated into groups of five for ease of comparison. In a few of the groups, the two sequences are identical; in others, there are one or more amino-acid differences. However, the similarity of the sequences of these distantly related species can be seen; both are variations on a common theme.

ditional musical training has such weight that one cannot help being stuck in that paradigm to some extent. Others have broken out of it—Subotnick comes to mind immediately—but I was not that confident of my own ability. So rather than study traditional music, I went to art school to study sound as art, and it was there that I discovered computers.

Digital computers have given electronic musicians new tools for developing deep structure. The computer's great strength is in its use as a compositional tool for algorithmic music—music that is developed with computer-processed rules that can be combined as tonal and structural relationships that would be difficult, if not impossible, to calculate by traditional means. Joseph Schillinger, who died in 1943, ironically the same year the first electronic computer was “born,” developed much of the groundwork for algorithmic music in his series of lectures that has been posthumously published as *The Schillinger Theory of Musical Composition* [3]. His theory that all music, perhaps all art, can be broken down to small whole-number ratios is difficult to align with traditional music composition techniques (although that is exactly what he attempts to do). However his theory is a perfect fit for computer algorithmic composition.

While algorithmic processes have given electronic art music a means of achieving deep structure, this structure is largely alien to our twentieth-century ears. And, since this music is still in the pioneer stage, with its paradigms still shifting and ephemeral, the listening audience for this kind of music remains negligible.

Thus, when botanist K.W. Bridges from the University of Hawaii asked me in 1989 to look into sonification of some of his data on tide tables, it occurred to me that, just as an artist's approach (rather than a musician's) helped loosen the bounds of tradition, perhaps substituting the structure of scientific data for the structure of cultural tradition would help lend form to electronic music that contemporary ears could appreciate.

While the tide-table data failed to resonate with any internal map I could discern, and the data were seemingly too random to give the resulting music a sense of structure, deep or otherwise, it led to discussions about what kind of scientific data might do this. Eventually the discussions with Bridges led me to look at DNA (deoxyribose nucleic acid) data and its associated protein sequences. It

... . - . - . - . - . - .
b e t a g l o b i n

Fig. 2. In this figure, the message “beta globin” is spelled out in Morse code. Each Morse codon consists of one or more dots and/or dashes. The individual codons are separated by a brief space (or silence in transmitted code) that allows the code reader to identify the individual letters of a message. Without this spacing, the combination “et” could not be distinguished from the “a” that follows it.

seemed to me that DNA's relatively simple alphabet of four coding elements that form just 20 “letters” (amino acids), which in turn combine to form the basis of all earth life, had to be rich with structure and very likely would resonate with the inner maps of humans, who are built upon this code.

This turned out to be the case. The DNA/protein sequences proved to possess deep and highly resonant structure that, when translated into music, sounds both alien and familiar, like music from another culture: pleasantly unusual but quite listenable. Our first public presentation of this music was in January 1981 at the University of Hawaii in a concert entitled “Inflections: Musical Interpretations of DNA Data,” which included music composed by myself and by Bridges along with related visuals performed by artist Sonia Sheridan.

At the time I thought the DNA/protein music would be a passing thing for me, a stepping stone on the exploratory search for compositional structure and meaning to parallel the remarkable electronic and digital tools that technology has given us. But the well has not run dry—how could it? Nature's music of life is on a far vaster scale than any human being (merely one of Her sonnets) could possibly surpass. But nature gives us a raw score so rich and harmonic it may well become the fountainhead for future sonic artists, just as She has provided for visual artists throughout human history.

As a research fellow in the arts at the University of Michigan for the past 2 years, I have collaborated with Jamy Sheridan, a visual algorithmic artist who has worked closely with me for several years on the algorithmic art and music software I have developed, and with Mary Anne Clark, co-author of this article. The collaboration with Clark began 2 years ago when she E-mailed me some technical questions regarding my software, which she had purchased. Further E-mail correspondence revealed that we were on similar trajectories regarding the sonification of protein data, but with two separate sets of keys: hers based on science and mine on art.

SONIFICATION OF DNA AND PROTEINS

DNA is a long, multi-unit molecule containing nature's digital code for life on earth. There are four coding elements: T, C, A and G. The letters stand for the four different subunits of DNA—thymine, cytosine, adenine and guanine—that form the “steps” on the helical ladder that is the database of all organisms. These four coding elements are combined into groups of three called codons. There are 64 possible codon combinations, of which 61 are used to encode the 20 amino acids and three are “stop” codons that indicate the end of a protein sequence, just as a period indicates the end of a sentence.

The 20 amino acids of which proteins are composed (see Table 1) differ from one another in size, solubility and electrical charge. Generally, water-insoluble amino acids such as leucine, isoleucine and valine cluster together in the interior of a protein, while more soluble amino acids are exposed on the surface. Positively charged amino acids such as lysine and arginine, and negatively charged amino acids such as glutamic and aspartic acid may also attract each other. These interactions encourage the protein to fold, like origami, into its functional form; the shape it assumes depends on the position of each amino acid in the sequence.

Just as a musical theme is defined not by the absolute pitches of the notes, but by the intervals from note to note, proteins are defined more by their overall patterns than by their absolute sequences. In order to form beta globin, the amino acids must line up in a way that allows the sequence to fold into a molecule capable of both binding and releasing oxygen depending on the physiological environment.

The amino acid interactions that stabilize a particular folding pattern must be preserved, even if the specific amino acid sequence is not, in order to preserve the function of a protein. The phrase (in amino acid letter names) FSDGL in human beta globin and the

[ATG]	GTG	CAC	CTG	ACT	CCT	GAG	GAG	AAG	TCT	GCC	GTT	ACT	GCC	CTG	TGG	GGC	AAG	GTG	AAC	GTG
[M]	V	H	L	T	P	E	E	K	S	A	V	T	A	L	W	G	K	V	N	V
	GAT	GAA	GTT	GGT	GGT	GAG	GCC	CTG	GGC	AGG	CTG	CTG	GTG	GTC	TAC	CCT	TGG	ACC	CAG	AGG
	D	E	V	G	G	E	A	L	G	R	L	L	V	V	Y	P	W	T	Q	R
	TTC	TTT	GAG	TCC	TTT	GGG	GAT	CTG	TCC	ACT	CCT	GAT	GCT	GTT	ATG	GGC	AAC	CCT	AAG	GTG
	F	F	E	S	F	G	D	L	S	T	P	D	A	V	M	G	N	P	K	V
	AAG	GCT	CAT	GGC	AAG	AAA	GTG	CTC	GGT	GCC	TTT	AGT	GAT	GGC	CTG	GCT	CAC	CTG	GAC	AAC
	K	A	H	G	K	K	V	L	G	A	F	S	D	G	L	A	H	L	D	N
	CTC	AAG	GGC	ACC	TTT	GCC	ACA	CTG	AGT	GAG	CTG	CAC	TGT	GAC	AAG	CTG	CAC	GTG	GAT	CCT
	L	K	G	T	F	A	T	L	S	E	L	H	C	D	K	L	H	V	D	P
	GAG	AAC	TTC	AGG	CTC	CTG	GGC	AAC	GTG	CTG	GTC	TGT	GTG	CTG	GCC	CAT	CAC	TTT	GGC	AAA
	E	N	F	R	L	L	G	N	V	L	V	C	V	L	A	H	H	F	G	K
	GAA	TTC	ACC	CCA	CCA	GTG	CAG	GCT	GCC	TAT	CAG	AAA	GTG	GTG	GCT	GGT	GTG	GCT	AAT	GCC
	E	F	T	P	P	V	Q	A	A	Y	Q	K	V	V	A	G	V	A	N	A
	CTG	GCC	CAC	AAG	TAT	CAC	TAA													
	L	A	H	K	Y	H	STOP													

Fig. 3. Genetic code for the protein beta globin [9]. These two sequences represent the amino-acid sequence of the human protein beta globin and the corresponding DNA sequence. The groups of three letters above the line represent the DNA codons. Below the line are the single-letter codes used for the 20 amino-acids. Each amino-acid is shown directly below its DNA codon. Although in this example the individual codons are separated by a space, the genetic code is read continuously, e.g. ATG GGT G CAC CTG ACT C C T G A G . . . In beta globin, the initial methionine (M) is removed from the final protein product. This sequence demonstrates the redundancy of the genetic code, even for a single protein. A given amino acid may be represented by any of several DNA codons. For example, lysine (K) is represented by both of its codons (AAA and AAG) and glycine (G) by three (GGG, GGC, GGT) of its four possible codons. The sequence also demonstrates how easily a variant can be introduced into the sequence. Altering the codon GAG to GTG would replace the first glutamic acid (E) of the sequence with valine (V). This single change produces the mutant beta globin of sickle-cell anemia.

phrase FGEAV in tuatara beta globin are different, but the amino acids at the last four positions of each cluster have similar charge and solubility characteristics. Such substitutions are said to be “conservative” and act a little like a musical key change, because they maintain the shape of the line even though the absolute sequence is changed.

HOW PROTEINS ARE ENCODED

Protein sequences and the organisms that contain them have the look of being designed or composed. The design of an organism and its molecular components emerges from the information stored in the DNA of its genes. The relationship between DNA coding sequences and protein structure is something like the relationship between Morse coding and plain text. Figure 2 demonstrates Morse code for the message “beta globin.” A

comparison of some features of the two coding systems—Morse code and genetic code—is useful:

- Morse code uses combinations of two elements, the dot and the dash, to specify letters of the alphabet and punctuation marks. In genetic code, combinations of the four subunits A, T, C and G are used to specify the 20 amino acids of the protein alphabet.
- Morse code uses coding combinations of various lengths—from a single dot (a short pulse) or dash (a longer pulse) to four dots/dashes—to represent the 26 letters of the English alphabet. Genetic code always uses combinations of the same size—three units. The DNA codons, e.g. AAA, CGA, CAT, specify the 20 amino acids, the alphabet of protein structure. Transmitted Morse code uses a brief period of silence to mark the boundaries between codons (e.g. to distinguish the letter combination “et” from

the letter “a” in the message “beta globin”). Genetic code is read continuously, parsing the DNA data string into triplets, and depends on the translating ribosomes, which match amino acids to RNA (ribonucleic acid) copies of the DNA codons, to get the reading frame right.

- Morse code begins with the first character of the message and uses a stop codon (.-.-) to specify the end of the message. Genetic code also begins with the first character of the message and ends with one of three stop codons: TAG, TAA or TGA. In both codes, the codons are laid out in the same sequence as the letters of the message.
- In Morse code, the relationship between codons and the letters of the message is fully unambiguous: either can be predicted from the other. V is only ...- and ...- is only V. However, genetic code is unambigu-

ous only when reading from the DNA to the protein. The reason that 61 DNA codons encode only 20 amino acids is that genetic coding is redundant. Most amino acids are represented by two or more codons (see Table 1 for a codon listing); only two amino acids are specified by a unique codon. Coding redundancy for several amino acids of a single protein can be seen in Fig. 3, which represents the DNA coding sequence and the corresponding amino acid sequence for human beta globin. For example, the amino acid lysine (K) is represented by both of its two DNA codons, sometimes by AAA and sometimes by AAG, and the amino acid glycine (G) is represented by three of its four possible codons—GGC, GGG and GGT.

These examples show that the sequence of a protein is not a fixed structure but a tentative one, like a melody in the mind of a composer. The theme played by a protein in one of its guises may turn up again as a variation or counter-theme in another part of the orchestra. In some cases, e.g. sickle-cell hemoglobin, a single amino acid substitution can seriously reduce the functionality of the protein. But sometimes a refolded tertiary structure develops new talents. The sickle-cell mutation has the side effect of increasing resistance to malaria. The normal beta globin is itself a variant of an earlier protein that also gave rise to other globins. Other protein variants have acquired completely new functions, e.g. the derivation of the milk protein lactalbumin from the protective enzyme lysozyme and the derivation of several eye-lens crystallins from respiratory enzymes [4].

The necessity for a working protein always to have some meaning, some function, has made proteins change slowly enough when they do change that they have left traces of their history behind in the record of their amino-acid sequences. Changes in protein sequences are generated by their “composers,” i.e. the DNA sequences that encode them. DNA produces new variations, both by making a change in the identity of a codon and by the wholesale recombination of themes taken from different DNAs. With the development of computer programs that can instruct digital musical instruments to play genetic scores, it has now become possible to hear these protein songs.

COLLABORATION OF ART AND SCIENCE

When we began the protein music project, we wanted to convey something both about primary amino acid sequence and about the folding patterns of proteins. Our goal was to create an audio CD that would stand on its own as art music and at the same time offer empirical proof of the aesthetic patterning of nature’s deep structure. One way to approach this was to take advantage of the secondary structure of proteins: simple folding patterns that are combined to produce the overall tertiary structure of a protein. There are three secondary patterns: alpha-helix, beta-strand and turns.

A protein chain is like a necklace, with the chemical groups that identify each amino acid dangling from the chain like pendants. These “pendants” are known as R-groups. The alpha-helix pattern looks like the binding of a “spiral” notebook or a strand of string wound at even intervals down a pencil. A helix is also like a spring in that it can be stretched along its long axis and, when released, will return to its original shape. In alpha-helix, the R-group “pendants” project outward from the axis of the helix.

Beta strands fold back and forth at the carbon atom to which the R-groups are attached. In beta strands, the R-groups project from the folded chain on alternate sides. Beta strands from different parts of the sequence or even from different sequences can line up with their R-groups in register. Adjacent strands form weak bonds that connect them into beta sheets or cylindrical beta barrels.

A turn is just that: a region of the molecule that goes off in a different direction from the one it came from. Turns may connect two regions of alpha-helix or beta strands to form alpha-turn-alpha or beta-turn-beta complexes, respectively, or they may connect alpha and beta regions.

As an element in the music that adds to its depth, the fact that these secondary structures exist in proteins, in addition to the variation and theme of the protein sequences themselves, is enough to make rich and interesting music. But to better understand how these simple patterns might contribute to even deeper structure in the music, we looked to the extra insight offered by the scientific study of more complex protein-folding patterns.

Various combinations of secondary structure form local domains in a

protein’s tertiary structure, or overall architecture. As cathedrals can be classified as Romanesque, High Gothic, Perpendicular and so forth, protein architectures are grouped into different categories, some of which are named simply for one of the proteins exhibiting the pattern, such as “immunoglobulin folds,” while others are named more descriptively, such as “trefoil (cloverleaf),” “Greek key” or “beta sandwich.”

The proteins we chose to work with in this project were representative of four major pattern categories: fibrous, predominantly alpha, predominantly beta and mixed alpha-beta. To distinguish between alpha and beta regions of these proteins and to mark the turns, we decided to use changes in instrumentation and/or pitch. For those proteins that have long regions in which one or more motifs are tandemly reiterated, we chose instead to use different voicings to differentiate between these motifs. What surprised us as we began to hear the sequences was that some of the alpha and beta regions also were marked by motifs whose sequences might not obviously repeat, but whose general shaping of phrases did.

DISCOVERING THE MUSIC IN PROTEINS

In Dunn’s previous music programs using DNA or protein sequences to generate music, he assigned pitches in one of two ways: absolutely, by giving a fixed pitch to each amino acid, or relatively, by making a frequency histogram of the amino acids in the protein and assigning more consonant intervals to the more frequently occurring amino acids. Because the properties of amino acids are important in determining folding pattern, we decided to recognize those properties by adding a third method for assigning pitch. We arranged the amino acids roughly according to their water solubility. The most insoluble residues were assigned pitches in the lowest octave; the most soluble, including the charged residues, were in the highest octave; and the moderately insoluble residues were given the middle range. Pitches ranged over three octaves for the diatonic scale, two octaves for a chromatic scale and about four octaves for pentatonic and whole-tone scales.

Since solubility scales are set according to various criteria about which there is no real consensus, we also paid attention to issues of harmony, setting the pitches

of amino acids with similar R groups at consonant intervals. Setting the scale according to solubility produced an interesting effect. As the linear sequence winds in and out of the interior of the protein, we hear counter-melodies in the music: one in the lower register representing the interior water-insoluble amino acids, and another in the upper register representing the more soluble amino acids arranged at the protein-water interface. Our linear sequences were playing two and sometimes three parallel and slightly offset tunes.

We also discovered another feature of the proteins: they had more than one personality. One of the earliest proteins that we set to music was lysozyme C, and it was set three times, twice by Dunn and once by Clark. This happened more or less by accident as we each prepared for lectures that we gave, along with visual artist Jamy Sheridan, at the Ann Arbor Museum of Art in May 1997. However, the experience of listening to these parallel compositions, each developed independently in two different locations (Clark in Texas, Dunn in Michigan), but with the same protein data and on the same sonification software, gave more insight into the astounding depth of structure Nature has built into Her art. Each piece was different from the others, so different that probably only someone very familiar with the lysozyme sequence would recognize it as the basis of the three pieces. We asked ourselves how the same sequence could assume these different characters.

One answer was relatively trivial: any piece assumes a different character if its rhythm, tempo and instrumentation are changed, just as the tune of "Amazing Grace" could function either as a march or as a lullaby, depending on such factors. The protein tunes also vary depending on which of the many pitch tables available to us are used. However, each of these tunes is an authentic voice of the protein, because of a critical feature of the proteins and nucleic acids as informational molecules. For each, there are so many possible combinations of tunes, it is often possible to specify a protein or DNA sequence uniquely by using fewer than 10 amino acids (or DNA codons) as the search pattern. This is not surprising, since the probability of any given sequence pattern of 10 amino acid residues randomly occurring would be $1/1.024 \times 10^{13}$. Indeed there are some combinations of 10 amino acids that do not appear in any protein now

recorded in the data bases. However, for a real protein, the pattern of pitch relationships produced by a given sequence will belong only to that protein, regardless of the pitch table used. Listening to a given protein's many voices is a way of inquiring into its nature, asking it, in poet Robert Frost's words, to "use language we can comprehend" [5]. And so, we interview each sequence many times, hoping to ask it the question that will produce an answer meaningful to us in terms of our own musical experience.

Because of the fruitfulness of multiple inquiry, we have continued to set individual proteins to music independently, as we did lysozyme, with Dunn asking "Where is the art in your science?" and Clark asking, "Where is the science in your art?" Our musical answers and the software we used to ask these questions are available on the Internet sites given below; we invite interested persons to add to the harmony with their own interpretations:

- John Dunn, Algorithmic Arts <<http://algoart.com>>
- John Dunn, DNA Music <<http://algoart.com/dnamusic>>
- M.A. Clark, The Music Room <<http://www.startext.net/homes/macclark/Music/musicpag.htm>>
- Kent (a.k.a. Kim) Bridges <<http://www.botany.hawaii.edu/faculty/bridges/>>.

APPENDIX: GENETIC MUSIC ANNOTATED SOURCE LIST BY M.A. CLARK

In his landmark book *Godel, Escher, Bach* (New York: Vintage Books, 1980; p. 519), Douglas Hofstadter comments on similarities between genes and music. The analogy is explicit in the following quote:

Imagine the mRNA to be like a long piece of magnetic recording tape, and the ribosome to be like a tape recorder. As the tape passes through the playing head of the recorder, it is "read" and converted into music, or other sounds. . . . When a "tape" of mRNA passes through the "playing head" of a ribosome, the "notes" produced are amino acids and the pieces of music they make up are proteins.

Hofstadter also discusses how meaning is constructed in protein and in music (p. 525):

Music is not a mere linear sequence of notes. Our minds perceive pieces of music on a level far higher than that. We

chunk notes into phrases, phrases into melodies, melodies into movements, and movements into full pieces. Similarly proteins only make sense when they act as chunked units. Although a primary structure carries all the information for the tertiary structure to be created, it still "feels" like less, for its potential is only realized when the tertiary structure is actually physically created.

The individuals and teams described below have taken advantage of the multiple biochemical and biophysical properties of both DNA and proteins to make the genetic patterns of these macromolecules audible. As Hofstadter first suggested, music is a natural medium for expressing the complex patterns of proteins and their encoding DNAs. Both consist of a linear sequence of elements whose real meaning lies in their combinations.

Hayashi and Munakata, using a system that assigned pitches to the four DNA bases according to their thermal stability within the interval of a fifth, found that converting the DNA sequences to music helped to expose the meaning of specific sequences and made remembering and recognizing specific DNA patterns easier. (Kenshi Hayashi and Nobuo Munakata, "Basically Musical," *Nature* **310** [12 July 1984] p. 96.)

Susumo Ohno, whose research explores the origin of life, proposed in 1986 that the meaning of proteins and of music springs from a similar origin—the repetition and elaboration of thematic sequences.

Ohno discusses the evidence that variations of two small primordial sequences—the decamer AAGGCTGCTG (= the peptide KAA) and a smaller derivative AAGCTG (= KL)—are reiterated again and again as primary themes in the sequences of genes, where they alternate with secondary themes composed of other sequences. To make the "repetitious recurrence" of these themes more vivid, Ohno developed a system of rules, based on the molecular weights of DNA's four bases, to convert the four bases into an octave scale. The system was used to produce a piece scored for violin: Human X-linked phosphoglycerate kinase. He also back-translated a Chopin nocturne into a DNA sequence that contains a remarkable 160-codon open-reading frame. Curiously, this sequence proved to be strikingly similar to the musical translation of the last exon of the gene for mouse RNA polymerase II. (Susumo Ohno and Midori Ohno, "The All Pervasive Principle of Repetitious Recurrence Governs Not Only Coding Sequence

Construction but also Human Endeavor in Musical Composition," *Immunogenetics* 24 [1986] pp. 71–78.)

In a later paper, Ohno explored another type of structure common to both music and protein sequences: the palindrome. In this paper, Ohno described the structure of mouse Histone H1, in which he found palindromic peptide sequences, some overlapping others, occupying 115 of the protein's 212 amino acids. The longest of these palindromes was the 15-residue sequence: KAVKPKAAKPKVAK. Using both the sequence of the functional protein and that of the "antisense protein" translated from the complementary strand of DNA, Ohno converted the Histone H1 sequence to music, using his previous developed pitch assignment rules, setting it as a piece that could be played either on piano or as an instrumental duet. (Susumo Ohno, "A Song in Praise of Peptide Palindromes," *Leukemia* 7, Supplement 2 [August 1993] pp. 5157–5159.)

David Deamer, another origin-of-life researcher, has also been intrigued by the musical properties of DNA. With composer Riley McLaughlin, he has produced two tapes (*DNA Suite* and *DNA Meditations*) of DNA music. Composer Susan Alexander and Deamer have also collaborated on the work *Sequencia*. Alexander describes this work in her web essay, Microcosmic Music—a New Level of Intensity. <<http://www.vll.com/SusanA/microcosmusic.html>>.

In *Sequencia*, pitches are assigned according to the light-absorption spectra of the four bases; the music uses a combination of synthesized tones and live instruments.

All of Alexander's works above can be obtained from: Science and the Arts, P.O. Box 8162, Berkeley, CA 94707, U.S.A., or from Susan Alexander: <<http://vll.com/SusanA/order.html>>.

Artist and programmer John Dunn (Algorithmic Arts) began creating and performing DNA-based music and developing genetic music software in 1991, first in collaboration with botanist K.W. Bridges and currently with biologist M. A. Clark. Dunn's software uses both frequency tables and amino acid characteristics (molecular weight, molecular volume and biochemical category) to assign pitch and other musical parameters to sequence data. Much of this music can be played from Dunn's Music from Life Web site at <<http://www.algoart.com/dnamusic/>>.

Another artist/scientist composing team is artist Peter Gena and medical geneticist Charles Strom, who presented their work demonstrating the translation of DNA sequences into music at the Sixth Symposium on Electronic Arts.

Gena and Strom use a sophisticated algorithm for converting the DNA sequences to music. Pitch is determined by a combination of the base composition of the codons and the dissociation constant of the amino acid encoded. Tone intensity is determined by the number of hydrogen bonds between base pairs, and duration of the tone by a combination of the dissociation constants and atomic weights of the amino acids. The amino acids encoded by each codon are also separated into eight chemical categories, with different instrumental timbres assigned to each. Thus nearly all musical elements of their pieces are determined directly by the codon sequence. (Peter Gena and Charles Strom, "Musical Synthesis of DNA Sequences," *Proceedings of the Sixth Symposium on Electronic Arts: Emergent Senses* [Montreal, Canada, 1995].)

Biologist Ross King and musician Colin Angus have collaborated to produce the piece "S2 Translation," (<http://www.nemeton.com/axis-mutatis/s2.html>), recorded on *The Shamen's CD Axis Mutatis* (<http://www.nemeton.com/axis-mutatis/index2.html>).

The DNA coding sequence for the S2 protein (a membrane receptor for the neurotransmitter serotonin) was converted to music that plays out both the DNA sequence and the sequence of the encoded protein. They assign the notes C, A, G and E to the bases cytosine, adenine, guanine and thymine (an interesting musical irony that recalls composer John Cage's assertion that music can be extracted from all the sounds around us). Under this melodic line, the bass progressions are structured to reflect the characteristics of the encoded amino acids, including their water-solubility, charge and size. Higher-order structure of the protein is suggested by changes in tonality. (The algorithm used in their software PM is described in their article Ross King and Colin Angus, "PM—Protein Music," *Computer Applications in the Biosciences* 12 [1996] pp. 251–252.)

Musical renditions of DNA and proteins are interesting not only as music, but as an alternative mode of studying genetic sequences. Humans have a keen ear for musical patterns, and using genetic sequences to generate music facili-

tates the recognition of those patterns. The rapid expansion of genetic databases driven in part by the Human Genome Project has made it clear just how much all life forms have in common. Similar genetic themes appear not only from species to species, but from protein to protein. Every genome is a study in the history of genetic composition.

The analytic and educational potential of using music to represent genetic patterns has been recognized from secondary school to university level. For example, Carol Miner and Paula Della Villa have developed a high-school learning project in which students create computer music from DNA sequences. (C. Miner and P. Della Villa, "DNA Music," *The Science Teacher* 64, No. 5 [1997] pp. 19–21.)

David Lane, the founder of Audio-Genetics (<http://www.audiogenetics.com/>), has built a DNA-based music company on the foundation of a project he began as a student at the University of Arizona. This ambitious enterprise recognizes and plans to develop many potential applications of DNA-based music to education, analysis and medicine.

My own (M.A. Clark) interest in using music to represent genetic patterns is also pedagogical. My collaboration with John Dunn, resulting in the *Life Music* CD described on the Protein Music Web site, began while I was teaching an honors course (Canons, Codons and Creativity, Marshall University, Spring 1996) on parallel patterns in genes and music. As a focal resource for this topic the class used Richard Powers' remarkable novel *The Gold Bug Variations* (New York: Harper, 1991), which also discusses similarities between genetic and musical coding.

Our goal for the *Life Music* CD was to represent not only the primary sequences of proteins, but the secondary folding patterns as well. Since alpha-helix, beta strands and turns each have characteristic combinations of hydrophobic and hydrophilic amino acids, different structural categories of proteins, which combine these secondary elements in different ways, also have different musical characteristics. The proteins we selected for the CD represent different protein types. The musical output of these and other proteins is available on the Protein Music Web site: <<http://www.startext.net/homes/macclark/Music/>>.

This source list also can be found at the Genetic Music web page: <<http://www.startext.net/homes/macclark/>>.

Music/Sources.htm>. For other links to algorithmic music sites, see the LINKS at Algorithmic Arts: <<http://www.algoart.com>>. Individuals who know of other resources for genetic music, have produced genetic music, or would like to suggest other additions or corrections to this listing, please contact M.A. Clark <macclark@startext.net>.

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